

Requirements:

- **Discuss** with your colleagues and help each other;
- Complete the report **independently**;
- The key of most following questions could be answered in a **mathematical** formulation. Therefore, try your best to give your mathematical answer instead of textural description;

1. Notations, variable, vectors, matrix, and others.
2. What means Laplacian?
3. What means eigenmap?
4. Introduce Laplacian Eigenmap in three steps. In each step, you should summary the assumptions, objective functions, limitations, and the optimization procedure (if existing), respectively. You must clearly distinct the concept/definition of motivation, assumption, problem and solution. These are important to understand any one paper.
5. Why the embedding objective function of LE is in its form? i.e., explain the physical meaning of each item in the objective functions and why the object function makes sense.
6. Giving the difference and connections among PCA, LLE, and LE from different aspects. (e.g. assumption, motivation, objective functions, and so on)
7. Experimental results on mnist (used in the previous reports).
 - a) You should perform LE on the training data and testing data separately.
 - b) After reducing the dim, perform KNN classifier ($k=1$) on the features to compute the classification accuracy.
 - c) You should find the user-specified parameters of LLE and understand their meanings? And then assigning different values to these parameters and report the corresponding classification accuracy;
 - d) Compare LE, LLE, and PCA on the classification task and 2-d visualization.
 - e) Design some experiments by yourself, and you should clearly give the designing principle (goal).